

### Indirect vaccine effectiveness of TCV

Indirect protection conferred by the Vi-TT typhoid conjugate vaccine has been assessed in a large cluster-randomised trial in Dhaka, Bangladesh. Conventional analysis and a subsequent “fried-egg” re-analysis both found modest, non-significant IVE of 17–19% despite high total VE (85%) and overall VE (57%) in the same study (1,2).

For the primary modelling scenario we therefore used IVE = 17%. To test the impact of more optimistic assumptions, we explored a bor

conducted a sensitivity analysis with IVE = 50%. This upper bound is informed by (i) the algebraic relationship, which yields  $\approx 50\%$  at the trial’s overall population coverage ( $f \approx 0.20$ ), and (ii) the 44 % indirect protection observed for the Vi-polysaccharide vaccine in Kolkata (3,4).

$IVE = (OVE - f \cdot TVE) / (1 - f)$ , where

OVE: overall vaccine effectiveness

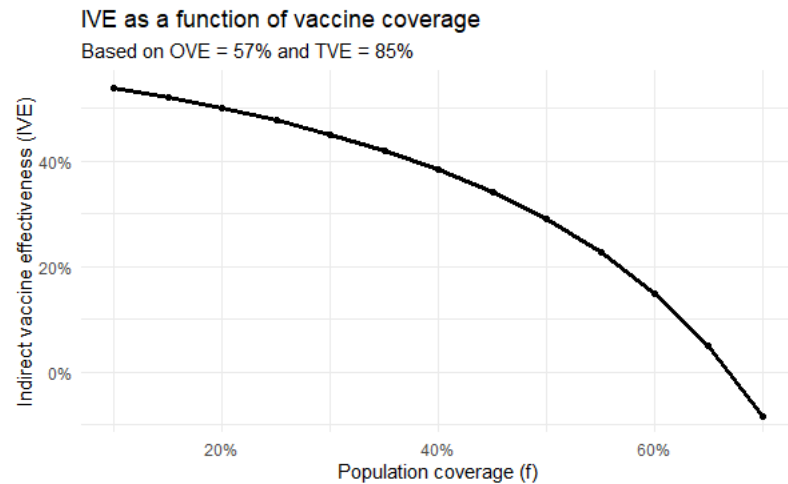
TVE: total vaccine effectiveness

IVE: indirect vaccine effectiveness

f: vaccine coverage

| Paper                        | Population size | Total vaccinated | Vaccine coverage (%) | Overall VE (%) | Total VE (%) | Indirect VE (%) | Ref |
|------------------------------|-----------------|------------------|----------------------|----------------|--------------|-----------------|-----|
| Ali 2020 (Vi-polysaccharide) | 31075           | 18869            | 61%                  | 57 (37-71)     | 61 (41-75)   | 44 (2-69)       | (3) |
| Khanam 2023 (Vi-TT)          | 155841          | 30882            | 20%                  | 57 (44-66)     | 85 (76-90)   | 17 (-13-40)     | (1) |

The Dhaka trial vaccinated 61 756 children among 311 289 residents, giving  $\approx 20\%$  population coverage. The theoretical IVE at this coverage ( $\approx 50\%$ ) exceeds the observed value, highlighting the limitations of the mass-action assumption in real-world settings with age-restricted vaccination, substantial migration, and mixed environmental and person-to-person transmission. Further field studies that include adults or achieve  $\geq 60\%$  population-wide coverage are needed to quantify the true potential for Vi-TT herd effects.



1. Khanam F, Kim DR, Liu X, Voysey M, Pitzer VE, Zaman K, et al. Assessment of vaccine herd protection in a cluster-randomised trial of Vi conjugate vaccine against typhoid fever: results of further analysis. *eClinicalMedicine* [Internet]. 2023 Apr 1 [cited 2025 Jul 23];58. Available from: [https://www.thelancet.com/journals/eclinm/article/PIIS2589-5370\(23\)00102-5/fulltext](https://www.thelancet.com/journals/eclinm/article/PIIS2589-5370(23)00102-5/fulltext)
2. Qadri F, Khanam F, Liu X, Theiss-Nyland K, Biswas PK, Bhuiyan AI, et al. Protection by vaccination of children against typhoid fever with a Vi-tetanus toxoid conjugate vaccine in urban Bangladesh: a cluster-randomised trial. *The Lancet*. 2021 Aug 21;398(10301):675–84.
3. Ali M, Sur D, Kanungo S, Qadri F, Kim DR, Islam T, et al. Re-evaluating herd protection by Vi typhoid vaccine in a cluster randomized trial. *International Health*. 2020 Jan 1;12(1):36–42.